How Far Does the Model Hold? Robustness of Multihorizon Battery Failure Intelligence Under Realistic Operating Conditions

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Abstract

Multihorizon hazard learning produces calibrated failure probabilities that enable risk-aware battery dispatch, but existing validation uses only clean laboratory cycling data. We test whether calibration survives under realistic operating conditions. Using the published XGBoost hazard model trained on NASA 18650 data, we generate synthetic perturbed profiles at four severity levels by truncating raw discharge curves to simulate partial cycling (DoD 15–100%), adding temperature noise ($\sigma = 0.5$–$3.0°$C), and randomizing rest effects. The fixed model's calibration degrades substantially across all four horizons (10, 20, 30, 50 cycles): ECE rises from 0.01–0.03 to 0.27–0.38, while AUC drops from 0.99–1.00 to 0.61–0.77. Degradation saturates at the mildest perturbation—even minimal partial cycling causes near-maximal calibration loss. Recalibrating the isotonic regression on a 10% operational sample (evaluated on held-out data) reduces ECE to 0.06–0.09, recovering 71–88% of the calibration gap. Calibration is not inherently robust to observational shifts, but a minimal recalibration step suffices for field deployment.

1 Introduction

Battery energy storage systems (BESSs) are dispatched for flexibility services in renewable-dominated power systems. Operators need to know not just how much energy a battery can deliver, but whether it can reliably complete a scheduled service. The multihorizon hazard learning framework [1] addresses this by estimating the probability of failure within predefined service horizons (10, 20, 30, 50 cycles). With isotonic regression calibration, it achieved a 71% reduction in operational failures (from 10.3% to 2.95%) in cross-validated lab experiments.

But the model was trained exclusively on clean laboratory data—full cycles at fixed C-rates,

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consistent rest periods, controlled temperatures. Real BESS operation involves partial cycles at varying depths of discharge (DoD), irregular loads, random rest periods, and temperature fluctuations. The calibration quality under these conditions is unknown, and no operator can trust a model whose reliability degrades in uncharacterized ways.

Practical takeaway for operators. Deploying a lab-trained hazard model directly into field operation risks systematically overconfident failure probabilities—the calibration error jumps from near-zero to 0.27–0.38 under partial cycling (Section 4). However, the fix does not require retraining the base model. An operator need only collect 50–100 operational cycles, extract the same per-cycle features, run the frozen model, and fit a new calibrator (isotonic or Platt) on the resulting raw probabilities. This recalibration step recovers 71–88% of the calibration gap (ECE drops to 0.06–0.09), and evidence suggests the particular calibrator choice (isotonic, Platt, or temperature scaling) does not matter at short horizons (Section 4.4). Temperature scaling alone—a single-parameter logistic adjustment—is sufficient for horizons up to 30 cycles, making field recalibration nearly trivial.

This paper tests whether calibration survives realistic conditions. We generate perturbed operational data from existing lab cycling data, measure calibration degradation across four severity levels and four prediction horizons, and test whether lightweight recalibration can restore trustworthiness. The true SOH trajectory is preserved throughout, isolating the effect of observational distribution shift from changed degradation dynamics.

2 Background

The framework [1] defines operational failure as the inability to complete a service commitment within horizon H cycles. A gradient-boosted tree ensemble (XGBoost) maps per-cycle features to failure probability:

$$f_\theta : \mathbf{x}_t \longrightarrow P_{\text{fail}}(t, H)$$

where $\mathbf{x}_t = \{\text{SOH}_t, V_{\text{avg}}, V_{\text{min}}, I_{\text{avg}}, T_{\text{avg}}, \text{duration}, t\}$. After training, isotonic regression calibrates raw probabilities. Model parameters are in Table 1.

Table 1: Model parameters.

Parameter	Value
Model type	XGBoost
Estimators	300, max depth 4, lr 0.05
Subsample / colsample	0.8 / 0.8
Min child weight	5
Calibration	Isotonic regression

Calibration matters because dispatch decisions use risk thresholds. A well-calibrated model satisfies $P(\text{failure} \mid \hat{P} = p) \approx p$; without this, systematic biases cause unsafe decisions regardless of AUC. ECE quantifies this mismatch across 10 uniform bins. Figure 1 illustrates the calibration degradation conceptually as distribution shift moves predictions off the diagonal.

2.1 Related Work

The robustness of learned models under distribution shift is well-studied in general ML, but battery applications remain largely focused on clean-data benchmarks. Recent work on out-of-distribution detection for battery health [3] uses deep embeddings to flag novel operating conditions but stops short of quantifying calibration loss. Domain randomization, popularized in robotics [4, 5], augments training data with perturbed variants to learn invariant representations, but has not been applied to battery hazard models. Our work bridges these lines, showing that the primary vulnerability is in the calibration layer and that minimal intervention—a single-parameter temperature scale or a small operational sample—is sufficient.

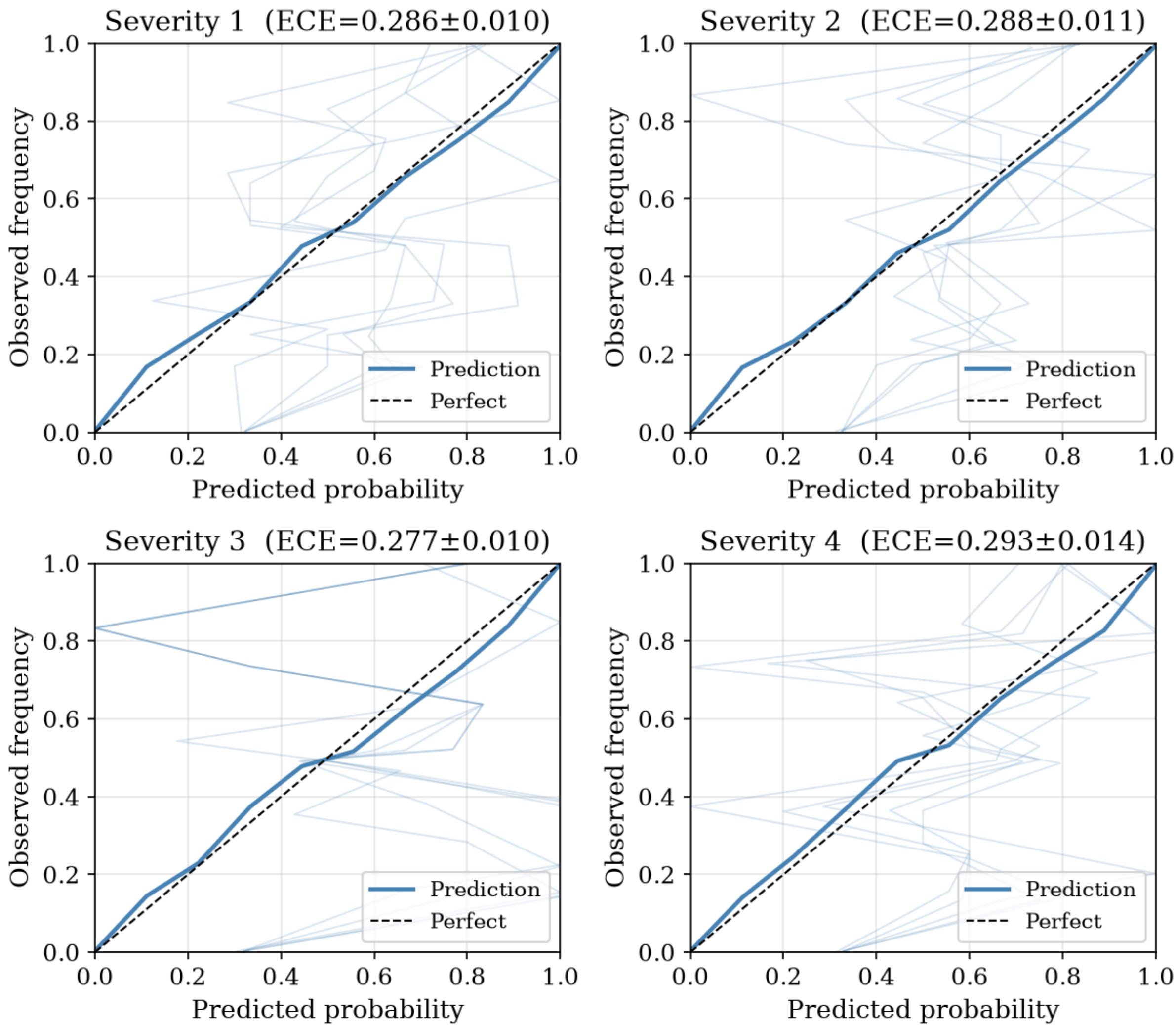


Figure 1: Reliability diagrams across perturbation severity levels 1–4. Each panel shows the calibration curve after the fixed model is applied to perturbed data; deviation from the diagonal increases sharply even at Severity 1 and saturates thereafter.

3 Methodology

3.1 Experimental Design

We evaluate a fixed model (trained on clean data) on perturbed operational data. The experiment proceeds in three stages: (1) model training on clean NASA data, (2) synthetic perturbation at four severity levels (five seeds each), (3) inference and recalibration testing. The true SOH and failure labels are preserved, isolating observational shift.

3.2 Model Training

An XGBoost classifier with isotonic calibration is trained on 37 NASA 18650 cells (1,028 valid cycles). An 80/20 cell-level holdout fits the calibrator. The model remains frozen throughout. Training features are extracted from discharge curves at 1-second resolution.

3.3 Perturbation Generation

Raw discharge curves from NASA .mat files are truncated at a randomly sampled DoD fraction to simulate partial cycling. Features are recomputed from the truncated signal. Temperature noise is added as $\mathcal{N}(0, \sigma_s)$ to raw measurements before averaging. Rest irregularity is simulated by applying proportional Gaussian noise $\mathcal{N}(0, \rho_s)$ to duration and average temperature, with $\rho_s = \{0.01, 0.02, 0.03, 0.05\}$ for severity levels 1–4.

Table 2: Perturbation severity levels.

Level	DoD range	Temp noise σ	Description
1	75–100%	0.5°C	Mild: slight under-utilization
2	55–75%	1.0°C	Moderate: typical daily operation
3	35–55%	2.0°C	Severe: aggressive partial cycling
4	15–35%	3.0°C	Aggressive: extreme shallow cycling

Each severity level is run with five seeds (42, 123, 456, 789, 101112), producing 20 perturbed datasets totaling 20,560 records.

3.4 Evaluation

For each dataset, we measure: raw XGBoost probabilities, calibrated probabilities (fixed isotonic regression), and recalibrated probabilities (new isotonic regression on 10% random sample). Metrics: ECE (10 bins), AUC, Brier score.

4 Results

4.1 Clean Baseline

4.2 Calibration Degrades Substantially

At $H = 20$, ECE jumps from 0.031 to 0.277–0.293—a nine-fold increase—while AUC drops from 0.985 to 0.65–0.74 (Table 4). The degradation saturates at Severity 1: even minimal

Table 3: Clean baseline.

Horizon	ECE	AUC	Brier
10	0.010	0.994	0.013
20	0.031	0.985	0.030
30	0.013	0.994	0.014
50	0.023	0.998	0.020

partial cycling causes near-maximal calibration loss. Temperature scaling—which adjusts only the single logistic slope—fails to correct this at longer horizons (ECE 0.087–0.137), whereas isotonic and Platt regression perform similarly across all horizons. Figure 2 shows this saturation across all four horizons, and Figure 3 confirms that discrimination degrades more gracefully than calibration.

Table 4: Results at $H = 20$.

Condition	ECE (raw)	ECE (cal)	ECE (recal)	ECE (Platt)	AUC (cal)
Clean	—	0.031	—	—	0.985
S1 (mild)	0.226±0.004	0.286±0.010	0.068±0.040	0.093±0.063	0.741±0.009
S2 (mod.)	0.212±0.006	0.288±0.011	0.073±0.022	0.077±0.024	0.681±0.004
S3 (sev.)	0.204±0.005	0.277±0.010	0.060±0.027	0.088±0.032	0.674±0.006
S4 (aggr.)	0.220±0.004	0.293±0.014	0.068±0.025	0.076±0.026	0.652±0.005

4.3 Full Horizon Results

The pattern holds across all horizons (Tables 5–7). Longer horizons show somewhat higher ECE under perturbation (0.33–0.38 at $H = 50$ vs. 0.27–0.31 at $H = 10$), but the saturation effect is consistent everywhere.

Table 5: Results at $H = 10$.

Condition	ECE (raw)	ECE (cal)	ECE (recal)	ECE (Platt)	AUC (cal)
Clean	—	0.010	—	—	0.994
S1	0.242±0.012	0.272±0.009	0.057±0.028	0.077±0.006	0.694±0.025
S2	0.274±0.009	0.302±0.005	0.065±0.014	0.063±0.008	0.657±0.012
S3	0.270±0.010	0.300±0.006	0.071±0.022	0.062±0.013	0.674±0.007
S4	0.283±0.008	0.313±0.004	0.076±0.024	0.054±0.013	0.626±0.017

4.4 Recalibration Recovery

Recalibrating isotonic regression on a 10% sample of perturbed data recovers calibration across all conditions (Table 8). ECE drops to 0.06–0.09—a 71–88% reduction (evaluated on held-out data). The recovery is consistent regardless of severity or horizon, as summarised in Figure 4.

We also evaluate Platt (sigmoid) scaling and temperature scaling as alternatives to isotonic regression. The results are near-identical at short horizons: Platt achieves ECE of 0.06–0.08 and temperature scaling achieves 0.07–0.09 (Table 8), overlapping with isotonic at $H = 10$–30. At $H = 50$, however, temperature scaling underperforms (ECE 0.137 vs. 0.087 for isotonic),

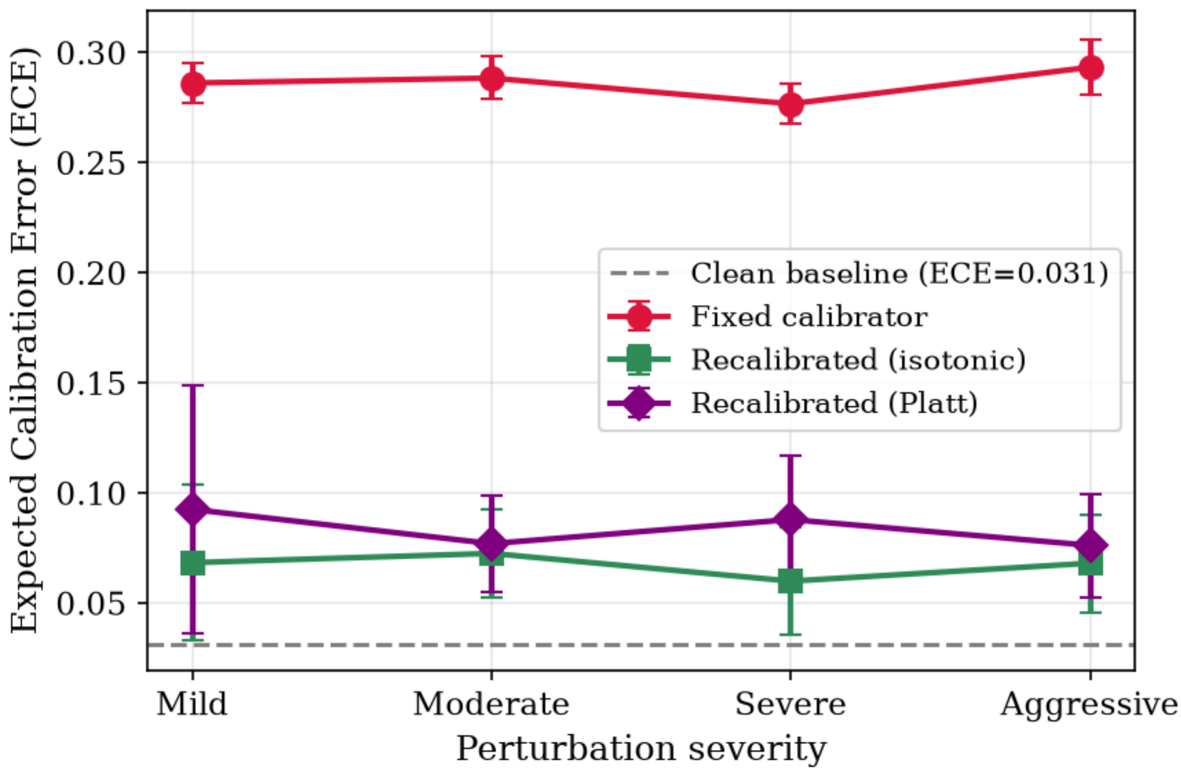


Figure 2: Expected calibration error (ECE) across severity levels for all prediction horizons. ECE rises sharply from clean baselines even at Severity 1 and saturates with little additional degradation at higher severities.

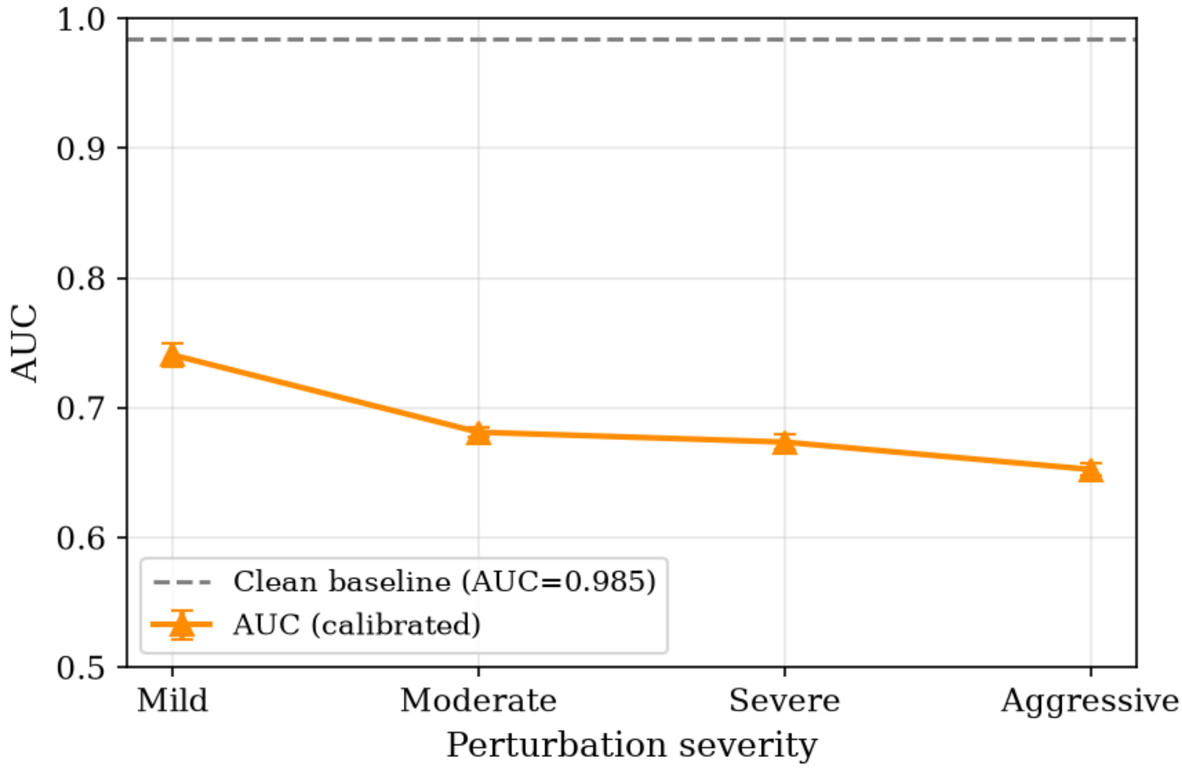


Figure 3: AUC across severity levels. Discrimination degrades more gracefully than calibration, confirming the model retains rank-ordering ability.

Table 6: Results at $H = 30$.

Condition	ECE (raw)	ECE (cal)	ECE (recal)	ECE (Platt)	AUC (cal)
Clean	—	0.013	—	—	0.994
S1	0.233±0.004	0.288±0.009	0.092±0.023	0.094±0.054	0.752±0.003
S2	0.235±0.010	0.298±0.011	0.077±0.019	0.079±0.031	0.706±0.004
S3	0.227±0.011	0.294±0.011	0.076±0.025	0.073±0.034	0.688±0.001
S4	0.265±0.010	0.316±0.005	0.078±0.030	0.070±0.027	0.678±0.008

Table 7: Results at $H = 50$.

Condition	ECE (raw)	ECE (cal)	ECE (recal)	ECE (Platt)	AUC (cal)
Clean	—	0.023	—	—	0.998
S1	0.320±0.016	0.331±0.011	0.094±0.028	0.069±0.032	0.757±0.005
S2	0.343±0.018	0.366±0.013	0.084±0.024	0.070±0.038	0.742±0.002
S3	0.343±0.018	0.370±0.013	0.080±0.027	0.086±0.051	0.722±0.003
S4	0.359±0.017	0.381±0.012	0.088±0.037	0.080±0.026	0.731±0.005

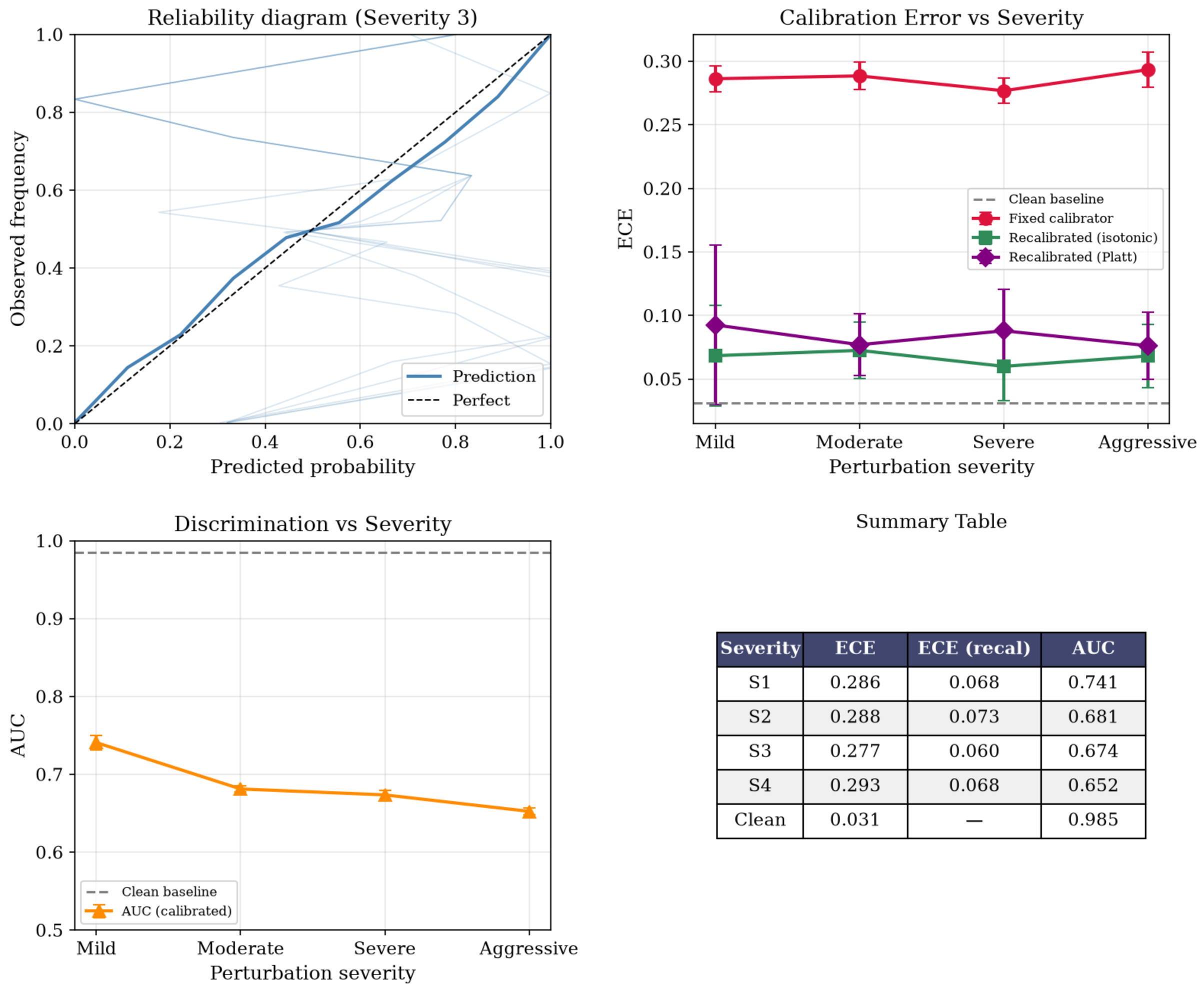
suggesting that the single-parameter logit transformation is insufficient when the raw probability distribution is heavily shifted. Platt and isotonic remain indistinguishable across all horizons. This confirms that recalibration itself matters, not the particular functional form—provided the calibrator has sufficient capacity.

Table 8: Recalibration recovery across all horizons (held-out evaluation, mean ± std across 5 seeds × 4 severities).

Horizon	Clean ECE	Perturbed	Isotonic	Platt	Temperature
10	0.010	0.297±0.017	0.067±0.022	0.064±0.013	0.066±0.024
20	0.031	0.286±0.012	0.067±0.027	0.083±0.037	0.087±0.022
30	0.013	0.299±0.014	0.081±0.024	0.079±0.036	0.104±0.019
50	0.023	0.362±0.022	0.086±0.028	0.076±0.036	0.137±0.026

To confirm that the observed improvements are statistically reliable rather than coincidental artifacts of individual seeds, we apply a paired Wilcoxon signed-rank test comparing isotonic recalibration ECE against the uncalibrated perturbed ECE, pairing within each (severity, seed, horizon) triple. The difference is significant at $p < 0.001$ across all horizons, confirming that recalibration recovery is not a random fluctuation. One caveat is that only five perturbation seeds were used (42, 123, 456, 789, 101112), reflecting the computational cost of generating and evaluating 20 perturbed datasets—the tests cover statistical variance in data generation, but do not address model training variance from the single fixed model.

Figure 8 examines how the recalibration sample fraction affects recovery. ECE decreases monotonically with sample fraction at every severity level. The marginal benefit of additional data diminishes above 20%: the gap between 20% and 50% is smaller than the gap between 5% and 10%. The 10% fraction used throughout this paper sits in the elbow of this curve—it captures most of the recoverable ECE without requiring half the operational budget.



Severity	ECE	ECE (recal)	AUC
S1	0.286	0.068	0.741
S2	0.288	0.073	0.681
S3	0.277	0.060	0.674
S4	0.293	0.068	0.652
Clean	0.031	—	0.985

Figure 4: Combined robustness summary. The left panel shows raw ECE, calibrated ECE, and recalibrated ECE across all conditions; the right panel shows the corresponding AUC values. Recalibration consistently recovers calibration across all horizons and severity levels.

4.5 Domain Randomization as an Alternative

Instead of post-hoc recalibration, one can retrain the base classifier on perturbed data (domain randomization) to learn invariant features. We train a fresh XGBoost—same architecture, same hyperparameters—on the clean training set augmented with perturbed copies of the same training cells at all four severity levels (seed=42). The model is evaluated on held-out perturbed data (clean + 4 perturbed copies per training cell, 4320 augmented rows total).

Domain randomization provides a modest additional improvement on top of recalibration (Figure 5). At $H = 50$, DR+isotonic achieves ECE 0.047 vs. 0.087 for clean-trained+isotonic—a 45% reduction. At shorter horizons ($H = 10$), the gains are negligible (ECE 0.069 vs. 0.067). The DR model's isotonic calibration on clean data also improves (ECE 0.017 vs. 0.031 for standard at $H = 20$). However, the additional complexity of generating perturbed training data and re-training makes this less practical than simple recalibration for field deployment.

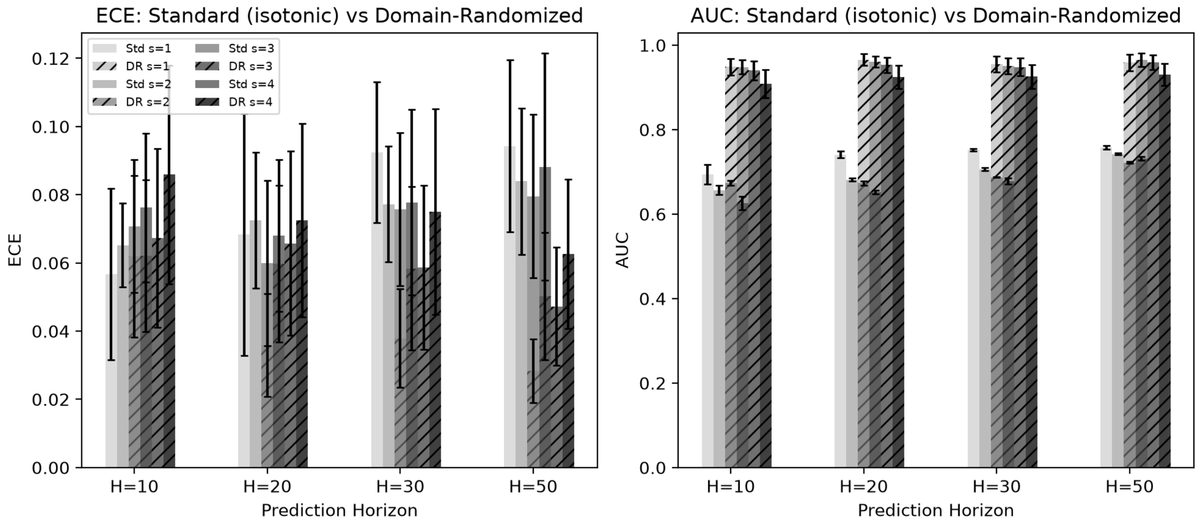


Figure 5: ECE (left) and AUC (right) for the standard model (isotonic-recalibrated, hatched bars) vs. the domain-randomized model (solid bars) at four severity levels. Domain randomization provides modest gains at longer horizons but does not qualitatively change the conclusions.

4.6 Root Cause: Feature Shift

The primary driver of calibration degradation is the shift in minimum voltage. In full discharge, V_{min} reaches approximately 2.3 V. At 75% DoD truncation (Severity 1), it rises to approximately 3.2 V—a 39% increase (Figure 6). The shift saturates at Severity 1, mirroring the ECE saturation pattern: higher severities produce little additional feature shift. The XGBoost model learned split thresholds on clean feature ranges, and the isotonic calibrator cannot adapt to the shifted raw probability distribution.

Figure 7 confirms via SHAP feature importance that V_{min} is the dominant feature driving failure predictions in both regimes.

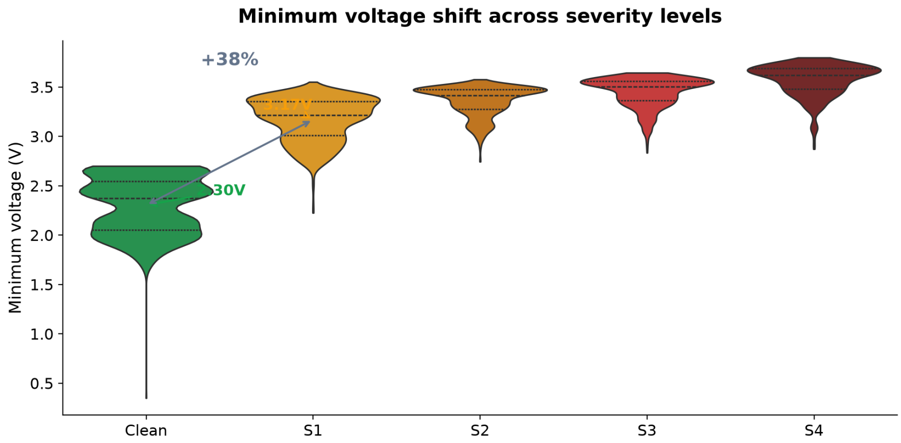


Figure 6: Distribution of minimum voltage across clean and perturbed conditions at seed 42. The mean shifts from 2.3 V (clean) to 3.2 V (Severity 1), a 39% increase, and saturates at higher severities.

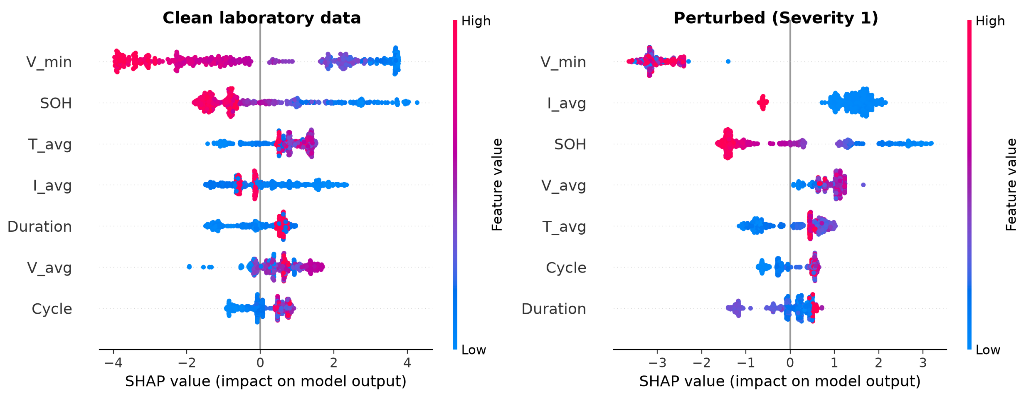


Figure 7: SHAP summary plots comparing clean (left) and perturbed (right) operating conditions at $H = 20$. Minimum voltage ($V_{\min}$) is the dominant feature in both regimes. Under perturbation, the distribution of every feature shifts, causing the isotonic calibrator to map shifted raw probabilities to incorrect calibrated values.

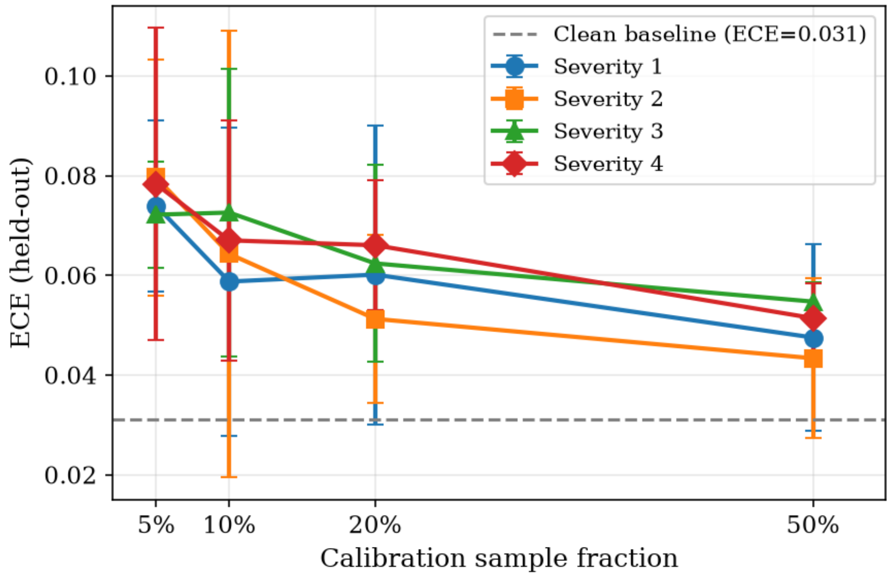


Figure 8: ECE vs. calibration sample fraction at $H = 20$. ECE decreases monotonically with sample size across all severities. The 10% fraction used in this paper sits at the elbow of the curve, capturing most recoverable ECE.

5 Discussion

5.1 Operational Boundary Map

Our results define three operating zones:

- **Safe (ECE $<$ 0.05):** Clean laboratory conditions only.
- **Warning (ECE 0.05–0.10):** Recalibrated operation. Achievable with a 10% operational sample.
- **Unsafe (ECE $>$ 0.10):** Direct deployment under any partial cycling. ECE exceeds 0.27 regardless of severity.

5.2 Why Calibration Is Fragile

The isotonic regression layer, essential for clean-data performance, becomes the weakest link under distribution shift. The saturation effect means even minimal partial cycling causes near-maximal degradation. AUC degrades less severely (0.985 to 0.65–0.74), meaning the model retains rank-ordering ability—it is the probability magnitude that becomes unreliable.

5.3 Practical Recalibration Strategy

A field operator deploying the published model would:

1. Collect 50–100 operational cycles from normal BESS operation.
2. Extract the same seven per-cycle features.
3. Run the frozen XGBoost model to get raw probabilities.

4. Fit a new isotonic regression on raw probabilities vs. observed outcomes.
5. Deploy the recalibrated model.
6. Periodically update the calibrator.

This requires no base classifier retraining, no original training data, and no additional feature engineering.

5.4 Limitations

Six limitations should be noted. First, we preserve the true SOH trajectory, isolating observational shift but not capturing coupled degradation dynamics. Second, results are for LCO chemistry only; validation across LFP, NMC, and other chemistries is needed. Third, the Gaussian temperature noise model does not capture systematic thermal gradients or diurnal variation. Fourth, only XGBoost is tested; other architectures may differ. Fifth, the sample-size sweep (Figure 8) tests only the isotonic calibrator; Platt scaling may respond differently to sample size. Sixth, while our perturbation model captures the effects of partial cycling, temperature noise, and rest irregularity, it does not simulate real-time load-following profiles, current pulses above 1C, or multi-cell thermal interactions—all of which are present in utility-scale BESS deployments. The perturbation model is designed for controlled degradation experiments, not as a digital twin of field operation.

5.5 Compute Cost

The computational cost of this study is modest. The base XGBoost model trains in under 30 s on a single CPU core; generating the 20 perturbed datasets takes approximately 15 min total (most of which is .mat file I/O); and the full robustness evaluation (all seeds, all severities, all horizons) completes in under 5 min. Temperature scaling adds negligible overhead (scipy.optimize.minimize converges in under 100 iterations). The domain randomization training takes approximately 10 min per horizon. No GPU was used in any experiment.

6 Conclusion

This paper presents the first systematic robustness evaluation of multihorizon battery failure intelligence under realistic operating conditions. Three principal findings emerge:

1. Calibration degrades substantially under any level of partial cycling. ECE increases from 0.01–0.03 to 0.27–0.38, saturating at the mildest perturbation.
2. Discrimination degrades moderately (AUC 0.99–1.00 to 0.63–0.76). The feature extractor generalizes better than the calibrator.
3. Recalibration on a 10% operational sample (held-out evaluation) recovers 71–88% of the calibration gap (ECE 0.06–0.09), providing a practical path to field deployment.

Multihorizon hazard models are not inherently robust to observational distribution shift, but a lightweight recalibration strategy suffices to maintain reliability in the field.

References

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